Design of reconfigurable assembly lines

The factory F manufactures a variety of equipment for the automobile sector. The assembly of each item necessitates multiple resources organized into distinct groups, where each resource group comprises a set of workstations.

Within a resource group, every workstation performs identical operations, although the equipment in each station could display minor variations. For instance, a resource group's workstations might place components on a base. Each station is equipped with a robotic arm and various tools. While the robotic arms in different stations are similar (e.g., they are all 5-axis robots), they might come from different brands (e.g., a brand "A" arm in the first station and a brand "B" arm in the second) and have different characteristics (e.g., different speed). An illustration of such an assembly line is presented in Figure 1, where three resource groups are displayed. Resource group A encompasses three stations, each featuring a robotic arm. Yet, the equipment in each station differs; the initial station uses an orange robotic arm, while the subsequent station employs a black one.

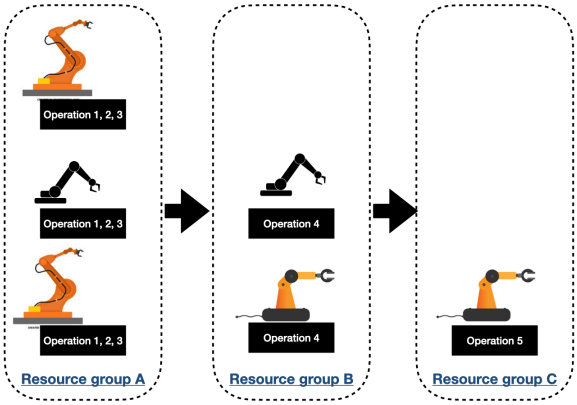


Figure 1: Example of a resource group

The factory owner needs help to decide how many stations to open in each resource group, which equipment to acquire, and how to assign the equipment among the resource groups. For each piece of equipment e, we are given a purchasing cost p_e, an operating cost b_{et} in period t, and a salvage price when the assembly line is dismantled. The operating cost accounts for the energy cost, average downtime of the machine, and maintenance cost.

The assembly line produces a single item, and the line is paced. That is, an item must come out of each resource group at every takt. For instance, each resource group is expected to produce an item every 30 seconds. To accommodate operation durations potentially longer than the takt time, each resource group comprises multiple parallel stations. Operation duration is given for the standard main resources, and each resource's speed is indicated as a multiple of the standard resource's speed. If for some periods, the available equipment in a resource group does not allow for meeting the takt time, the factory owner faces a large cost associated with the missed opportunity. This penalty is computed based on the largest deviation from the takt among all stations. This penalty corresponds to some lost sales because a line that does not meet the takt time has a too small throughput to meet the demand.

In practice, assembly line design decisions are made in a highly uncertain environment. The item produced on the line evolves every year. We assume the changes in variant impact only the duration of the operation and not the required resources. The owner of the factory considers the provided estimated operation duration may very well vary by 100% of the estimated values. In this work, we will assume the pieces of equipment are bought in a period 1 and kept in the factory for the entire life cycle of the line (10 years). However, every year, the owner of the factory takes advantage of the Christmas break to reconfigure the line. During this period, he/she can open or close stations and move the equipment from one station to another.